MEMORY FORENSICS REPORT

MFF v2 · Post-Volatility Analysis Framework

OVERALL SEVERITY: CRITICAL

Case ID	`case01_baseline_vs_case02_t1055_5_attack`
Generated	`2026-03-12 00:20:22 UTC`
Baseline capture	`case01_baseline`
Attack capture	`case02_t1055_5_attack`
Baseline path	`/MFF/cases/case01_baseline`
Attack path	`/MFF/cases/case02_t1055_5_attack`

1. EXECUTIVE SUMMARY

Metric	Value	Metric	Value
New Processes (Attack Only)	13	Critical Risk Processes	2
Gone Processes (Baseline Only)	9	ATT&CK Techniques Observed	1
New Network Connections	131	ATT&CK Tactics Observed	1
Flagged Connections	3	IOCs Extracted	42

ATT&CK; Tactics Observed:

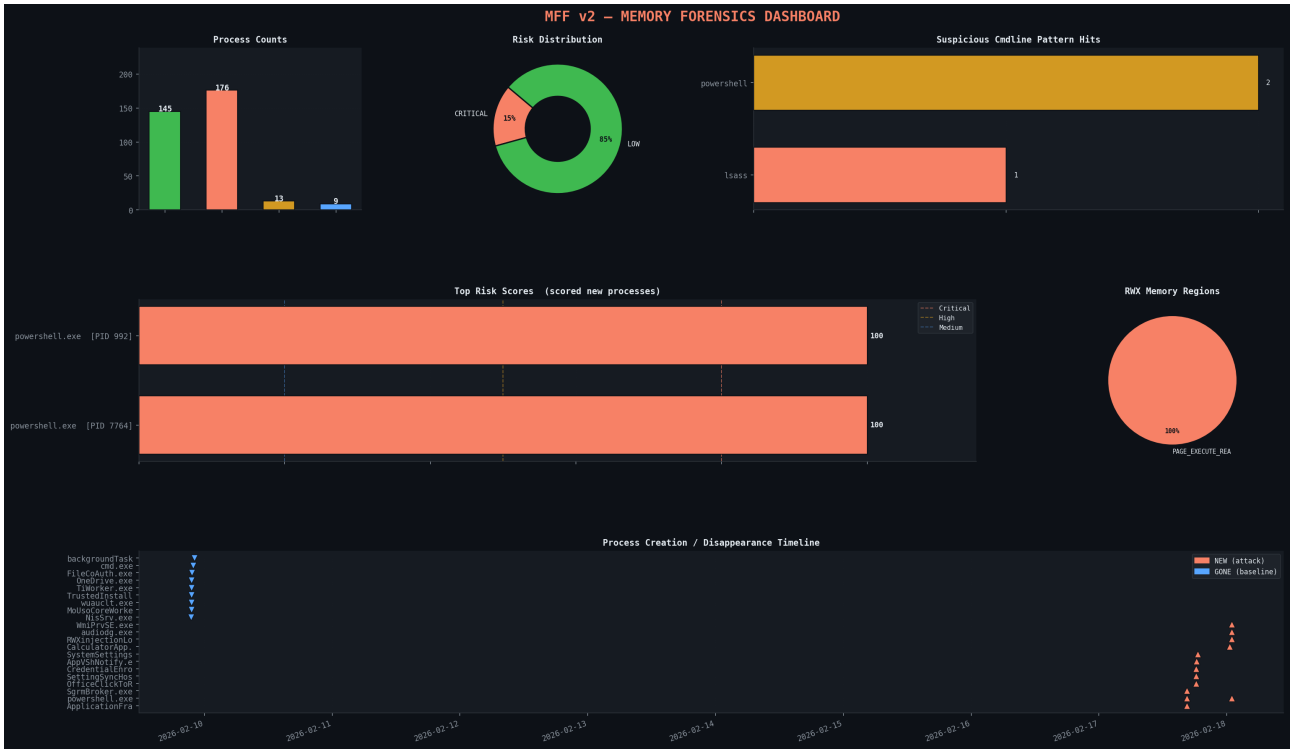
Execution

IOC Summary:

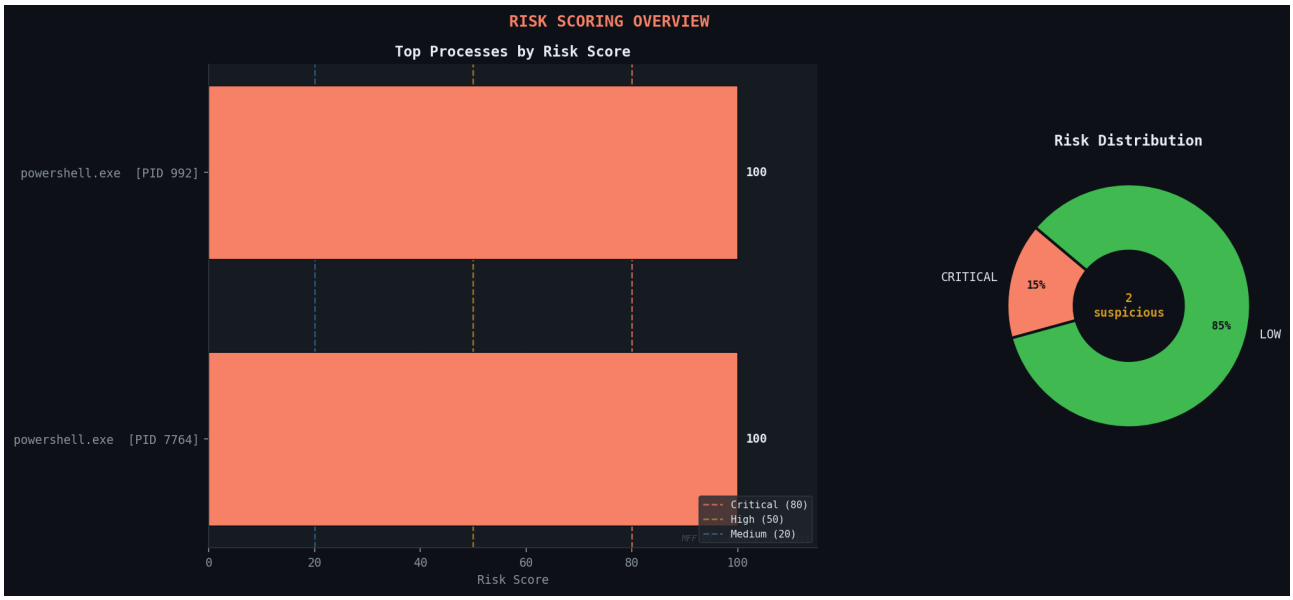
Type	Indicators (top 8)
IPv4	0.0.0.0 · 150.171.28.11

2. VISUALIZATIONS

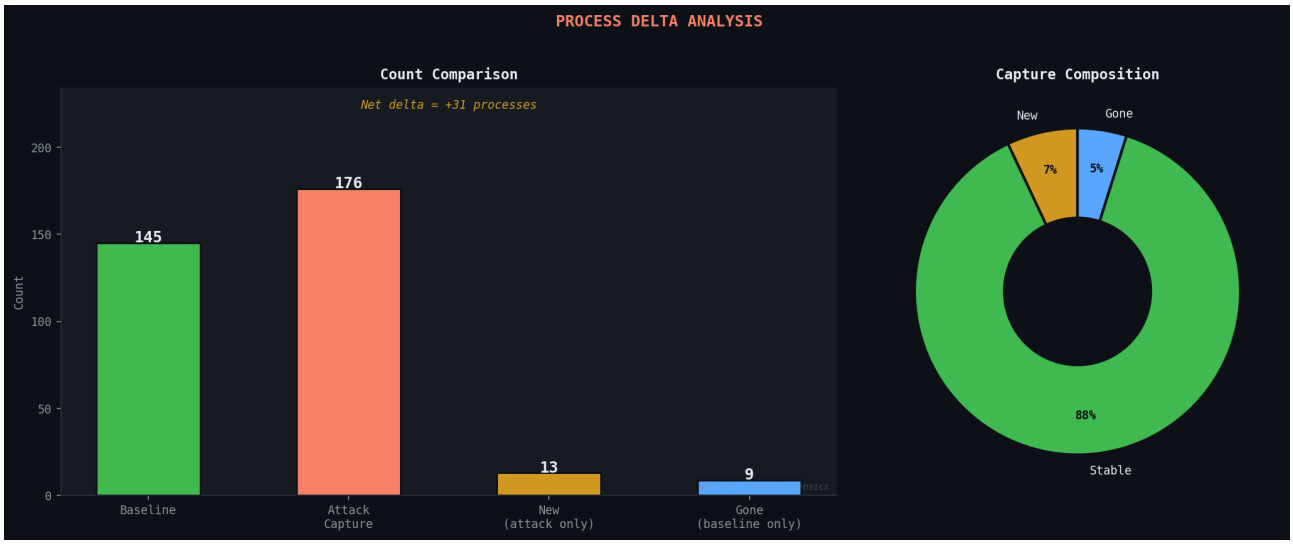
2.1 Full Forensics Dashboard



2.2 Risk Scoring Overview

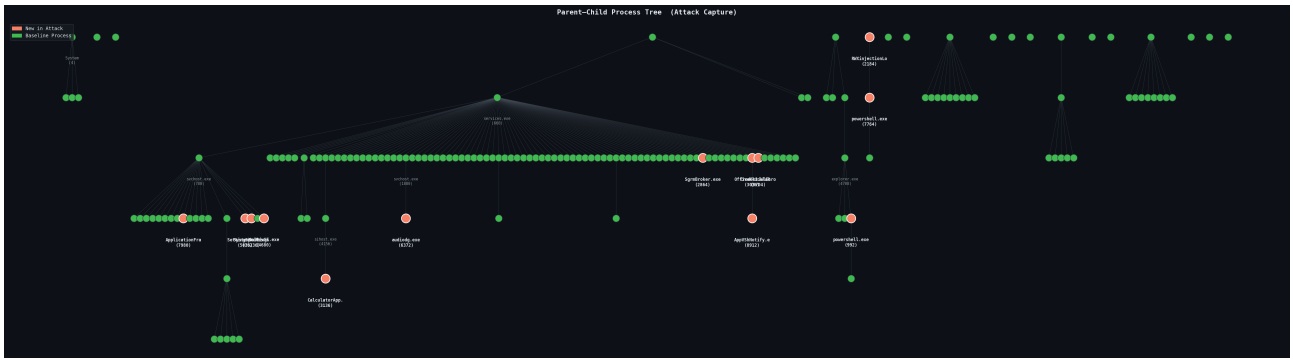


2.3 Process Delta Analysis

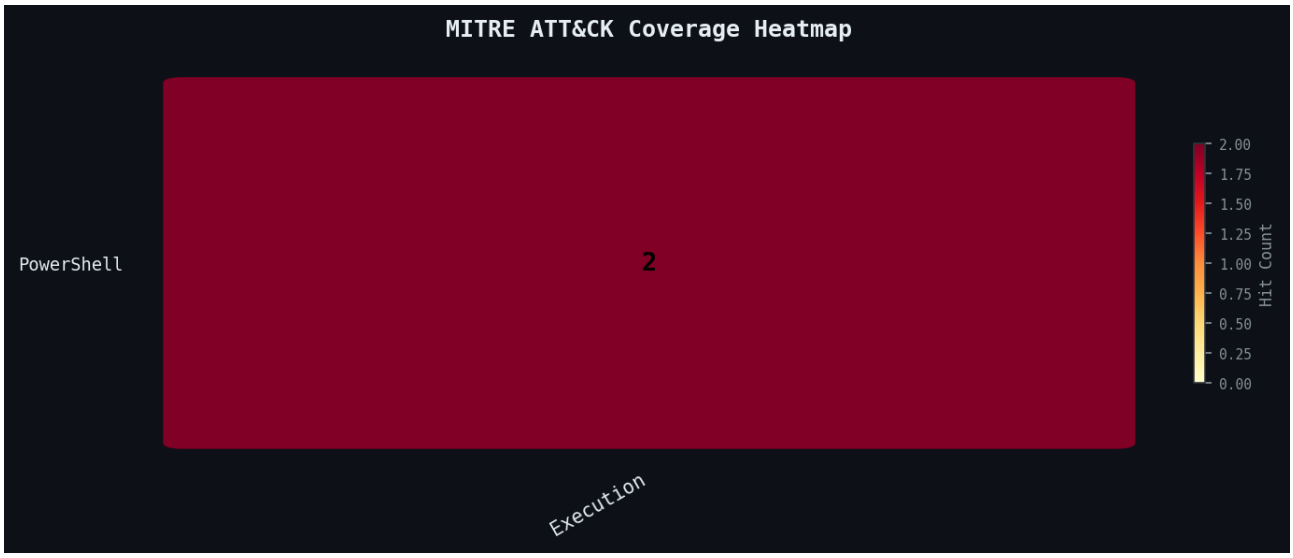


2.4 Process Timeline

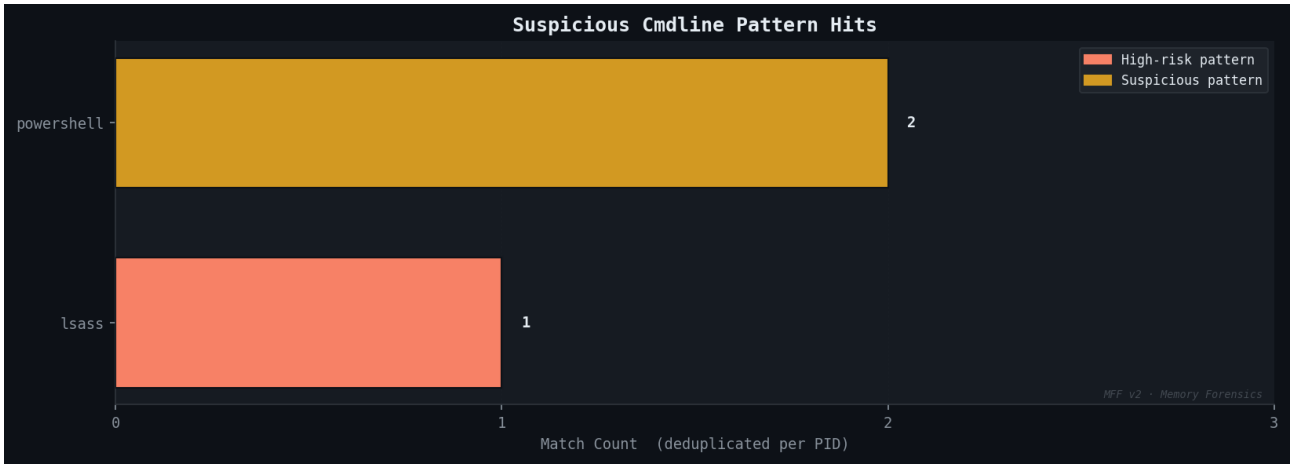
2.5 Parent-Child Process Tree



2.6 MITRE ATT&CK; Heatmap



2.7 Suspicious Cmdline Patterns



2.8 Malfind Memory Protections

3. RISK SCORE TABLE

Process	PID	Score	Level	Indicators
powershell.exe	992	100	CRITICAL	Suspicious cmdline pattern; RWX executable memory region (malfind)
powershell.exe	7764	100	CRITICAL	Suspicious cmdline pattern; RWX executable memory region (malfind)

4. NEW PROCESSES (Attack Only)

ImageFileName	PID	PPID	CreateTime
ApplicationFra	7980	780	2026-02-17 16:30:03.000000 UTC
powershell.exe	992	4708	2026-02-17 16:30:29.000000 UTC
SgrmBroker.exe	2864	660	2026-02-17 16:30:58.000000 UTC
OfficeClickToR	3016	660	2026-02-17 18:12:01.000000 UTC
SettingSyncHos	5076	780	2026-02-17 18:14:10.000000 UTC
CredentialEnro	9704	660	2026-02-17 18:14:21.000000 UTC
AppVShNotify.e	8912	3016	2026-02-17 18:18:11.000000 UTC
SystemSettings	10236	780	2026-02-17 18:32:00.000000 UTC
CalculatorApp.	3136	4156	2026-02-18 00:31:14.000000 UTC
RWXinjectionLo	2184	9076	2026-02-18 00:55:10.000000 UTC
powershell.exe	7764	2184	2026-02-18 00:55:10.000000 UTC
audiodg.exe	6372	1880	2026-02-18 00:55:11.000000 UTC
WmiPrvSE.exe	4680	780	2026-02-18 00:55:16.000000 UTC

5. MITRE ATT&CK; TECHNIQUE MATCHES

Technique	TechniqueName	Tactic	MatchedKeyword	PID	Process
T1059.001	PowerShell	Execution	powershell	992	powershell.exe
T1059.001	PowerShell	Execution	powershell	7764	powershell.exe

6. SUSPICIOUS CMDLINE FINDINGS

PID	Matched Pattern	Args
992	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
7764	powershell	\\??\\C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -Command "Add-Type -AssemblyName PresentationFramew
680	lsass	C:\\Windows\\system32\\lsass.exe

7. MALFIND — MEMORY INJECTION ARTEFACTS

Process	PID	Protection
powershell.exe	992	PAGE_EXECUTE_READWRITE
powershell.exe	992	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE

8. NETWORK ARTEFACTS

8a. Flagged Connections (suspicious ports)

TreeDepth	Offset	Proto	LocalAddr	LocalPort	ForeignAddr	ForeignPort	State	PID	Owner	Created	SuspiciousPort	PortMeaning
0	0xdb84c1572230	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	900	svchost.exe	2026-02-17 15:28:51.000000 UTC	135	DCOM/RPC
0	0xdb84c1572230	TCPv6	::	135	::	0	LISTENING	900	svchost.exe	2026-02-17 15:28:51.000000 UTC	135	DCOM/RPC
0	0xdb84c1572e90	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	900	svchost.exe	2026-02-17 15:28:51.000000 UTC	135	DCOM/RPC

9. IOC DETAIL TABLE

Type	Value	Source	Count
FilePath	C:\\Windows\\system32\\svchost.exe	cmdline	46
FilePath	C:\\Windows\\System32\\svchost.exe	cmdline	29
FilePath	C:\\Program	cmdline	9
FilePath	C:\\Windows\\System32\\RuntimeBroker.exe	cmdline	6
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView	cmdline	4
FilePath	Microsoft\\EdgeWebView\\Application\\144.0.3719.115\\msedgewebview2.exe	cmdline	4
FilePath	??\\C:\\Windows\\system32\\conhost.exe	cmdline	2
FilePath	C:\\ProgramData\\Microsoft\\Windows	cmdline	2
FilePath	C:\\Windows\\system32\\DllHost.exe	cmdline	2
FilePath	Microsoft\\Edge\\Application\\msedge.exe	cmdline	2
FilePath	Windows	cmdline	2
FilePath	system32\\csrss.exe	cmdline	2
FilePath	.....	malfind	1
FilePath	.....\\.....	malfind	1

Type	Value	Source	Count
FilePath	??\\C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView\\Crashpad	cmdline	1
FilePath	C:\\Windows\\Explorer.EXE	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxService.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxTray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe	cmdline	1
FilePath	C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\spoolsv.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\\StartMenuExperienceHost.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\\TextInputHost.exe	cmdline	1
FilePath	C:\\Windows\\system32\\AUDIODG.EXE	cmdline	1
FilePath	C:\\Windows\\system32\\ApplicationFrameHost.exe	cmdline	1
FilePath	C:\\Windows\\system32\\SearchIndexer.exe	cmdline	1
FilePath	C:\\Windows\\system32\\SecurityHealthService.exe	cmdline	1
FilePath	C:\\Windows\\system32\\SgrmBroker.exe	cmdline	1
FilePath	C:\\Windows\\system32\\lsass.exe	cmdline	1

10. METHODOLOGY

Dataset: Memory images captured using VMware hypervisor snapshot of Windows 10 VM. Attack simulations performed with Atomic Red Team (ART) scripts.

Volatility 3: Memory parsed using Volatility 3 plugins: pslist, cmdline, malfind, netscan, pstree. Output exported as CSV.

Process diff: Baseline vs attack comparison by process name (ImageFileName). PID-based diff rejected — PIDs reassigned on every reboot.

Risk scoring: Transparent rule-based scoring (no ML). RWX memory region +60 · Suspicious cmdline +40 · Known-malicious name +50 · Anomaly bonus +15.

Malfind filter: Known-clean JIT processes excluded (browsers, Defender, .NET). Only genuinely suspicious RWX regions retained.

ATT&CK; mapping: 25+ rule-based signatures across 8 tactics. Keyword matching on cmdline args, process names, and network artefacts.

IOC extraction: Regex-based extraction: IPv4 (excluding RFC1918), domains, URLs, MD5/SHA1/SHA256, Windows file paths.

Framework: MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP project. Tool chain: Volatility 3 → MFF v2 → HTML/PDF reports.